

LAB 3: COMPONENT MODELLING & ARCHITECTURAL PATTERN SELECTION

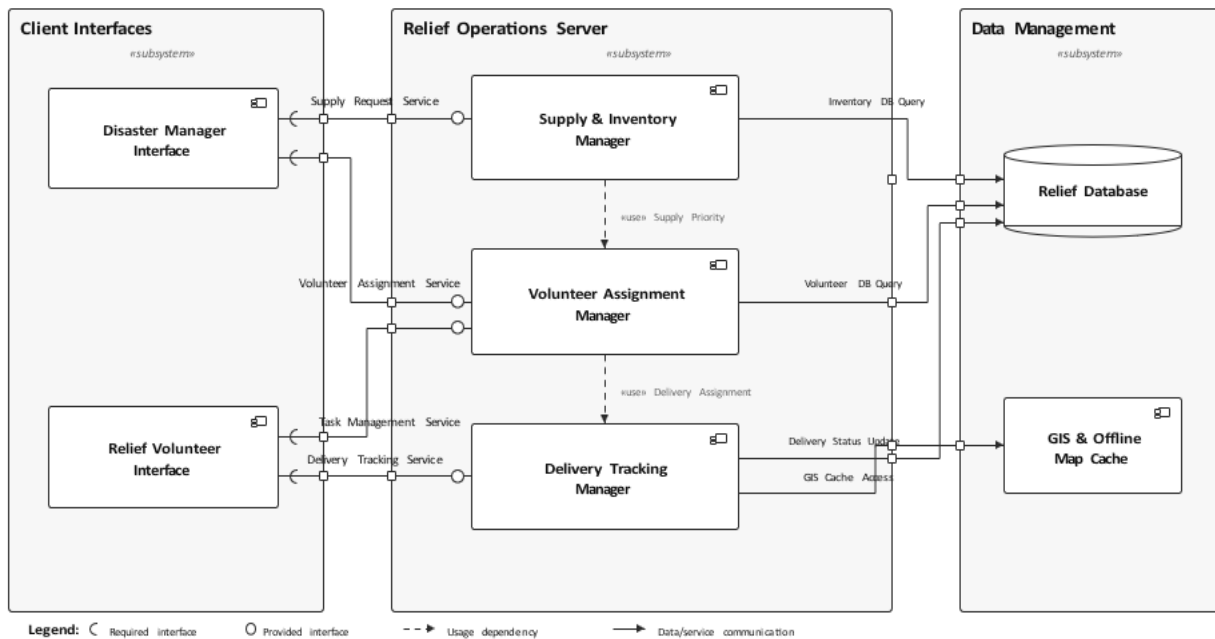
Problem Statement #64 - Disaster Relief Supply & Volunteer Coordinator

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COMPONENT DAIGRAM:



1. Architectural Style Analysis

Style	Strength for this system	Limitation for this system
Layered	Clear separation of user interface, business logic and data access.	Does not directly emphasize communication between remote Disaster Manager/Volunteer clients and shared central services.
Microservices	Independent services can scale and isolate failures.	Adds deployment, network-latency and data consistency complexity, which is undesirable in low-bandwidth disaster conditions.
Client-Server	Centralized services and shared operational data are easy for both actors to access consistently.	The server must be designed carefully because it can become a bottleneck or single point of failure.

2. Architecture Selection

We chose Client-Server Architecture for the Disaster Relief Supply & Volunteer Coordinator.

Reason 1 - Shared operational data: Disaster Managers and Relief Volunteers need access to common shelter requests, warehouse inventory, volunteer assignments and delivery-status information. A centralized server provides one authoritative source of data and supports the Lab 1 reliability requirement that this information remain accurate during normal operation.

Reason 2 - Appropriate complexity: The system has a focused set of coordinated functions: supply management, volunteer assignment and delivery tracking. Client-server keeps these services centralized while allowing multiple remote clients, avoiding the extra operational complexity and network overhead of a microservices design.

Security Advantage: In the proposed design, authentication, role-based authorization and request validation can be enforced at the central server before a Disaster Manager or Relief Volunteer accesses operational services. This gives a single controlled point for protecting shared disaster-relief data.

Performance Benefit: The Lab 1 requirement calls for low-bandwidth operation and offline GIS data caching. The client-server design supports this by keeping the main operational data centralized while allowing GIS/map information to be cached locally on clients, reducing repeated network transfers when connectivity is weak.

Major Components: Disaster Manager Interface, Relief Volunteer Interface, Supply & Inventory Manager, Volunteer Assignment Manager, Delivery Tracking Manager, Relief Database, and GIS & Offline Map Cache. The component diagram shows four primary provided/required service interfaces plus database and GIS dependencies.